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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Master of Computer Applications

Semester – III

Data Analytics for Business

Final Project

Title: Creating BlinkIT Dashboards

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BlinkIT Grocery Sales Dashboard

Developing the BlinkIT Grocery Sales Dashboard

Github Link: [https://github.com/Priyanshnolkha/Data-Analytics-For-Business-Projects/tree/main/Final Project Blinkit Grocery Dashboard](https://github.com/Priyanshnolkha/Data-Analytics-For-Business-Projects/tree/main/Final%20Project%20Blinkit%20Grocery%20Dashboard)

Power BI Service Link: [View BlinkIT Dashboard in Power BI Service](#)

Problem Statement and Data Collection:

Given Problem Statement:

You are a Business Intelligence Consultant for a growing organization. Your task is to develop an end-to-end Business Analytics solution using a dataset of your choice to support data-driven business decisions.

Project-Specific Problem Statement:

BlinkIT's outlet and item-level sales data is spread across raw files with no consolidated view for management, making it difficult to track performance across outlet types, locations, and item categories. The task is to import this data into Power BI, clean and model it, and build an interactive dashboard that shows total sales, item counts, ratings, and outlet-wise performance to support quick, data-driven decisions.

Data Description:

Dataset used: BlinkIT Grocery sales data, sourced from a MySQL database (blinkit_db) and loaded into Power BI via Power Query. Each row represents one item sold at a specific outlet. Key columns include:

- **Item Fat Content** — Low Fat or Regular classification of the item.
- **Item Identifier** — unique code assigned to each item (e.g., FDY49, NCJ54).

- **Item Type** — product category such as Fruits and Vegetables, Snack Foods, Household, Dairy, Frozen Foods, Canned, Baking Goods, Health and Hygiene, Meat, Soft Drinks, Breads, Hard Drinks, Starchy Foods, Breakfast, and Seafood.
- **Outlet Establishment Year** — year the outlet was set up (used to build the establishment trend chart, 2012–2022).
- **Outlet Identifier** — unique code for each outlet (e.g., OUT027).
- **Outlet Location Type** — the outlet's location tier: Tier 1, Tier 2, or Tier 3.
- **Outlet Size** — Small, Medium, or High, describing the physical size of the outlet.
- **Outlet Type** — Grocery Store, Supermarket Type1, Supermarket Type2, or Supermarket Type3.
- **Sales, Item Visibility and Rating** — transactional measures used to compute Total Sale, Avg Sales, No. of Items, Item Visibility and Avg Rating.

Business Problem:

Retail and category-management teams need a fast way to see which outlets and item categories are actually driving sales — not just by raw revenue, but by rating, visibility and outlet format. This project builds a single interactive dashboard that answers that question at both the catalogue level and the individual-outlet level.

Data Cleaning & Transformation (Power Query):

The BlinkIT Grocery data was connected in Power BI via Get Data → MySQL Database, pointing to the `blinkit_db.blinkit_grocery` table. The following transformations were applied in Power Query Editor before loading to the model:

- Previewed the table in the Navigator window to confirm the columns — Item Fat Content, Item Identifier, Item Type, Outlet Establishment Year, Outlet Identifier, Outlet Location Type, and the remaining outlet/sales fields.
- Promoted Headers — the first row of the incoming MySQL table was promoted to become column headers.
- Changed Type — each column's data type was set correctly (text for identifiers/categories, whole number for Outlet Establishment Year, decimal for sales/rating/visibility measures).

- Verified there was a single query — "BlinkIT Grocery Data" — with no fully-blank rows before closing & loading to the model.

Screenshot Image:

Item Fat Content	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type
Regular	FDY49	Canned	2018	OUT027	Tier 3
Low Fat	NCJ54	Household	2018	OUT027	Tier 3
Low Fat	FDP11	Breads	2018	OUT027	Tier 3
Low Fat	FDR23	Breads	2018	OUT027	Tier 3
Low Fat	FDB27	Dairy	2018	OUT027	Tier 3
Low Fat	FDK51	Dairy	2018	OUT027	Tier 3
Low Fat	FDK28	Frozen Foods	2018	OUT027	Tier 3
Low Fat	FDK52	Frozen Foods	2018	OUT027	Tier 3
Low Fat	FDU52	Frozen Foods	2018	OUT027	Tier 3
Low Fat	NCK05	Health and Hygiene	2018	OUT027	Tier 3
Low Fat	NCA06	Household	2018	OUT027	Tier 3
Low Fat	NCE06	Household	2018	OUT027	Tier 3
Low Fat	NCG07	Household	2018	OUT027	Tier 3
Low Fat	NCH54	Household	2018	OUT027	Tier 3
Low Fat	NCT18	Household	2018	OUT027	Tier 3
Low Fat	NCZ06	Household	2018	OUT027	Tier 3
Low Fat	NCL07	Others	2018	OUT027	Tier 3
Regular	FDF24	Baking Goods	2018	OUT027	Tier 3
Regular	FDF26	Canned	2018	OUT027	Tier 3
Regular	FDR40	Frozen Foods	2018	OUT027	Tier 3
Regular	FDW16	Frozen Foods	2018	OUT027	Tier 3
Regular	FDY04	Frozen Foods	2018	OUT027	Tier 3
Regular	FDO31	Fruits and Vegetables	2018	OUT027	Tier 3

Screenshot Image:

Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type
FDY49	Canned	2018	OUT027	Tier 3
NCJ54	Household	2018	OUT027	Tier 3
FDP11	Breads	2018	OUT027	Tier 3
FDR23	Breads	2018	OUT027	Tier 3
FDB27	Dairy	2018	OUT027	Tier 3
FDK51	Dairy	2018	OUT027	Tier 3
FDK28	Frozen Foods	2018	OUT027	Tier 3
FDK52	Frozen Foods	2018	OUT027	Tier 3
FDU52	Frozen Foods	2018	OUT027	Tier 3
NCK05	Health and Hygiene	2018	OUT027	Tier 3
NCA06	Household	2018	OUT027	Tier 3
NCE06	Household	2018	OUT027	Tier 3
NCG07	Household	2018	OUT027	Tier 3
NCH54	Household	2018	OUT027	Tier 3
NCT18	Household	2018	OUT027	Tier 3
NCZ06	Household	2018	OUT027	Tier 3
NCL07	Others	2018	OUT027	Tier 3
FDF24	Baking Goods	2018	OUT027	Tier 3
FDF26	Canned	2018	OUT027	Tier 3
FDR40	Frozen Foods	2018	OUT027	Tier 3
FDW16	Frozen Foods	2018	OUT027	Tier 3
FDY04	Frozen Foods	2018	OUT027	Tier 3
FDO31	Fruits and Vegetables	2018	OUT027	Tier 3

Data Modelling:

The model uses a single fact table built directly from the cleaned MySQL source:

- BlinkIT Grocery Data (fact table) — one row per item sold at an outlet, holding Item Fat Content, Item Type, Outlet Establishment Year, Outlet Location Type, Outlet Size, Outlet Type, Sales, Item Visibility and Rating.
- No additional dimension tables were required — all slicers (Outlet Location Type, Outlet Size, Item Type) and visuals filter directly off this single table, keeping the model simple and performant.

DAX Measures and KPIs:

1. Metrics Field Parameter:

DAX Measure: metrics = { ("Total Sale", NAMEOF('BlinkIT Grocery Data'[Total Sale]), 0), ("Avg Sales", NAMEOF('BlinkIT Grocery Data'[Avg Sales]), 1), ("No Of Items", NAMEOF('BlinkIT Grocery Data'[No Of Items]), 2), ("Avg Rating", NAMEOF('BlinkIT Grocery Data'[Avg Rating]), 3) }

This field parameter powers the four toggle buttons above the Fat Content / Item Type visuals, letting the same donut and bar charts swap between measures interactively instead of needing four separate sets of visuals.

Screenshot Image:

```
1 metrics = {  
2     ("Total Sale", NAMEOF('BlinkIT Grocery Data'[Total Sale]), 0),  
3     ("Avg Sales", NAMEOF('BlinkIT Grocery Data'[Avg Sales]), 1),  
4     ("No Of Items", NAMEOF('BlinkIT Grocery Data'[No Of Items]), 2),  
5     ("Avg Rating", NAMEOF('BlinkIT Grocery Data'[Avg Rating]), 3)  
6 }
```

2. Total Sales:

- Total Sales = \$1.20M

3. Avg Sales:

- Avg Sales = \$141

4. No Of Items:

- No Of Items = 8,523

5. Avg Rating:

- Avg Rating = 3.9

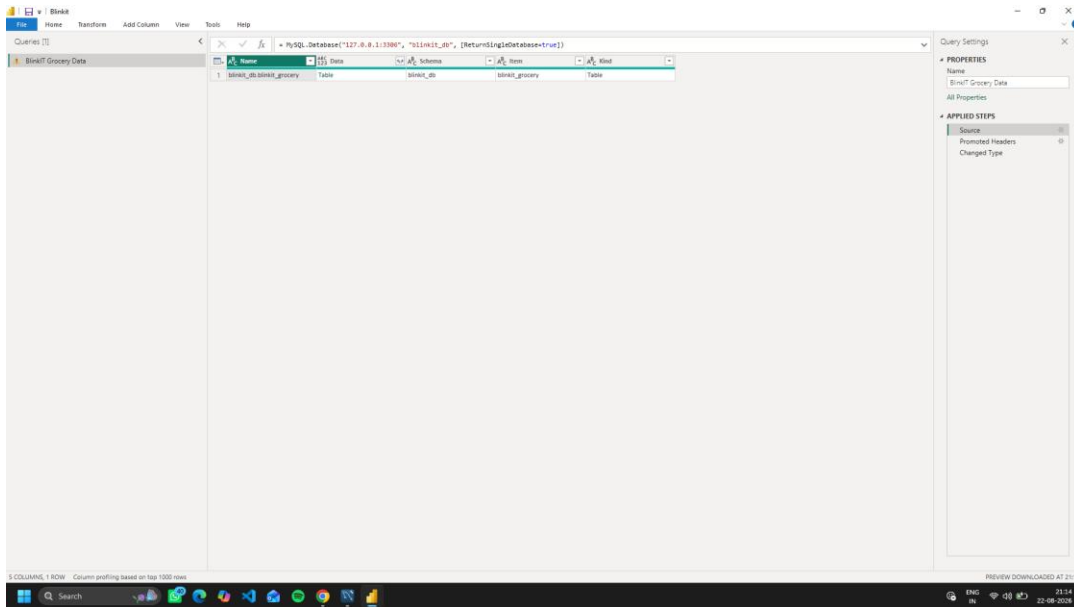
(The KPI cards above are visible together in the dashboard screenshot in the Interactive Dashboard section below.)

MySQL Server Connection With Powerbi Desktop:

- Connected to the MySQL server at 127.0.0.1:3306 using Get Data → MySQL Database in Power BI Desktop.
- Selected the blinkit_db database with ReturnSingleDatabase set to true, and loaded the blinkit_grocery table.
- Renamed the query to "BlinkIT Grocery Data" and connected it to the existing Power BI data model.
- Enabled the dashboard to use database data with refreshable connections.

```
= MySQL.Database("127.0.0.1:3306", "blinkit_db", [ReturnSingleDatabase=true])
```

Screenshot Image:



Methodology:

All KPI cards, charts, and slicers were arranged into a single interactive BlinkIT Grocery Sales dashboard in Power BI. The dashboard includes:

- Total Sales KPI Card
- Avg Sales KPI Card
- No. of Items KPI Card
- Avg Rating KPI Card
- Outlet Establishment Trend (Line Chart, 2012–2022)
- Total Sale by Fat Content (Donut Chart)
- Fat Content by Outlet Location Tier (Bar Chart)
- Total Sale by Item Type (Bar Chart)
- Outlet Size Distribution (Donut Chart)
- Total Sale by Outlet Location Tier (Bar Chart)
- Detailed Outlet Type Table (Total Sale, No. of Items, Avg Sales, Avg Rating, Item Visibility)
- Interactive Slicers: Outlet Location Type, Outlet Size, and Item Type

- Toggle buttons to switch the main view between Total Sale, Avg Sales, No. of Items, and Avg Rating

The dashboard enables management to analyze sales performance across outlet types, locations, and item categories, monitor item visibility and ratings, and identify high- and low-performing outlets, supporting data-driven retail decision-making.

Interactive BlinkIT Dashboard Image:

- BlinkIT Grocery Sales Dashboard provides an interactive overview of outlet and item-level sales performance.
- Displays key metrics such as Total Sales, Avg Sales, No. of Items, and Avg Rating.
- Shows the Outlet Establishment trend from 2012 to 2022, peaking at \$205K around 2018.
- Analyzes Fat Content, Item Type, Outlet Size, and Outlet Location Tier contributions to total sales.
- Includes a detailed Outlet Type table with Total Sale, No. of Items, Avg Sales, Avg Rating and Item Visibility.
- Interactive slicers and toggle buttons allow users to explore the data and gain meaningful insights.

Screenshot Image:



Power BI Service Workspace:

- Published the BlinkIT Grocery Sales dashboard to the "Blinkit Dashboard" workspace in Power BI Service for online access and sharing.
- The dashboard is available in the workspace with interactive KPIs, charts, filters, and the same toggle-button field parameter as the desktop version.

The published report can be accessed through the following Power BI Service link:

[View BlinkIT Dashboard in Power BI Service](#)

- The Power BI Service enables the dashboard to be viewed, shared, and accessed online from the workspace.

Screenshot Image:



Business Question Answered:

I. What is the overall sales performance across all BlinkIT outlets?

Answer: Across all outlets, BlinkIT generated \$1.20M in total sales from 8,523 items, averaging \$141 in sales per item and a customer rating of 3.9 out of 5 — indicating healthy, consistent performance.

II. Which year of outlet establishment produced the strongest sales performance?

Answer: Outlets established around 2018 perform strongest, with the establishment trend peaking at \$205K that year, up from \$78K in 2012, before easing to \$131K by 2022.

III. Do Regular or Low Fat items drive more sales, and does this hold across outlet tiers?

Answer: Regular items outsell Low Fat items overall (\$776.32K vs \$425.36K) and this pattern holds consistently across Tier 1, Tier 2, and Tier 3 outlets.

IV. Which item categories should be prioritised for shelf space and promotion?

Answer: Fruits and Vegetables and Snack Foods are the strongest categories at \$0.18M each, while Seafood (\$0.01M) and Breakfast (\$0.02M) contribute the least and are candidates for review.

V. Which outlet size and location tier contribute the most to sales?

Answer: Small-format outlets contribute the most (\$507.90K), ahead of High (\$444.79K) and Medium (\$248.99K) outlets; by location, Tier 3 outlets lead with \$472.13K, followed closely by Tier 2 (\$393.15K) and Tier 1 (\$336.40K).

VI. Which outlet type is the strongest overall performer, and are there any outliers worth investigating?

Answer: Supermarket Type1 is the clear leader with \$787.55K in sales from 5,577 items. Grocery Store outlets are a notable outlier: despite carrying only 1,083 items, they match the 4-star average rating of larger formats and post the highest item visibility (0.10), suggesting an efficient, well-curated assortment.

Business Insights and Recommendation:

Key Insights:

- Sales are concentrated in a small number of item categories — Fruits and Vegetables and Snack Foods together account for a disproportionate share of revenue compared to categories like Seafood and Breakfast.
- Regular (non-low-fat) items consistently outperform Low Fat items across every outlet tier, suggesting fat content is a strong, stable driver of purchase behaviour rather than a location-specific trend.
- Outlets established in 2018 significantly outperform both earlier and later cohorts, hinting at a specific expansion strategy or store format from that year worth replicating.
- Small-format outlets outperform Medium outlets on total sales despite the assumption that larger stores drive more revenue, indicating strong per-outlet efficiency in the Small segment.
- Grocery Store outlets achieve the highest item visibility (0.10) with comparatively few items (1,083), showing that a focused assortment can still deliver strong customer engagement.

Recommendations:

- Expand shelf space and promotional focus on Fruits and Vegetables and Snack Foods, while reviewing the assortment strategy for consistently weak categories like Seafood and Breakfast.
- Investigate the operating model of outlets established around 2018 (format, location strategy, staffing) to identify replicable success factors for future outlet openings.
- Study the Small-format outlet playbook, since these outlets generate the highest total sales despite their size, and consider applying similar assortment/space strategies to Medium outlets.
- Use the Grocery Store outlet type's high item-visibility, low-item-count model as a benchmark for improving product discoverability at Supermarket Type2 and Type3 outlets, which post the lowest visibility (0.06).
- Continue tracking Avg Rating alongside Total Sale using the field-parameter toggle, so that any outlet with rising sales but falling ratings can be flagged early.

Conclusion:

This project delivers an end-to-end Business Intelligence solution for BlinkIT's grocery retail operations: item and outlet-level data was connected live from a MySQL database, cleaned and typed in Power Query, and modelled into a single "BlinkIT Grocery Data" table enriched with a DAX field parameter that drives interactive Total Sale / Avg Sales / No Of Items / Avg Rating toggle views. The resulting interactive Power BI dashboard — published to both Power BI Desktop and Power BI Service — lets management drill from the full \$1.20M, 8,523-item catalogue down to a single outlet type, location tier, or item category in a few clicks, surfacing insights that can directly inform assortment planning, outlet expansion strategy, and retail decision-making.